Project: Dataset Visualization

Group 7
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Business Understanding and Data

Business objective

Build an interactive dashboard to explore video game sales and show differences between markets.

Main questions

- Which platforms and genres perform best in different regions?
- How have sales and platform popularity changed over time?
- Which games have the highest reported worldwide sales?
- Which publishers and platforms show strong sales performance?

Dataset overview

- Main dataset: `vg_charts.csv`
- **64,016 rows and 14 columns** in the supplied CSV.
- Key attributes: title, platform, genre, publisher, developer, critic score, regional sales and release date.
- Supporting datasets include developer, publisher and geographic information.

Data Preparation

How we prepared the data

1. **Remove duplicates** using `drop_duplicates()`.
2. **Clean text columns** by removing surrounding whitespace and converting empty strings to missing values.
3. **Convert dates** using `pd.to_datetime()`.
4. **Convert sales and critic scores** to numeric data types using `pd.to_numeric()`.
5. **Create `release_year`** from `release_date` to support time-based analysis.

- Values that cause errors are converted to NaN

```python
# Rensar vg_charts dataset och skapar en relase_year kolumn
def clean_charts(df):
    df = df.copy()

    # Tar bort dupbletter
    df = df.drop_duplicates()

    # Tar bort alla withespaces på kolmerna som ska vara av datatypen string
    text_columns = ["img", "title", "platform", "genre", "publisher", "developer"]
    for column in text_columns:
        if column in df.columns:
            df[column] = df[column].astype("string").str.strip()
            df[column] = df[column].replace("", pd.NA)

    # Konventerar de värden som ska vara Datetime till rätt datatyp
    # Värden som inte kan omvandlas till ett datum blir NaT (Not a Time)
    for column in ["release_date", "last_update"]:
        if column in df.columns:
            df[column] = pd.to_datetime(df[column], errors="coerce")

    # Omvandlar kolumnerna till numeriska värden och sätter ogiltiga värden till NaN
    numeric_columns = ["critic_score", "total_sales", "na_sales", "jp_sales", "pal_sales", "other_sales"]
    for column in numeric_columns:
        if column in df.columns:
            df[column] = pd.to_numeric(df[column], errors="coerce")

    # Skapar en ny kolumn med release_year från release_date kolumn
    # ex 2026-10-15 blir 2026 samt värden som inte går att konventeras blir Na
    df["release_year"] = df["release_date"].dt.year.astype("Int64")

    return df
```

Interactive Dashboard and Visualization

From data to insights

Our Dash application uses interactive visualizations to explore:

- Regional sales differences.
- Platform and genre popularity.
- Sales trends by release year.
- Top-selling games and publisher performance.

Visualization choices

- **Bar charts:** Compare platforms, genres and games.
- **Line charts:** Examine changes in sales and popularity over time.
- **Interactive filters:** Focus on particular platforms, regions, genres or years where supported by the application.
- **Tooltips:** Inspect individual values without overcrowding the charts.

Live Demo

Key Findings

1. Regional differences

- North America has the largest sum of reported regional sales
- Sports is the leading genre by reported sales in North America.
- Role-playing games lead by reported genre sales in Japan.
- Action games narrowly lead Sports in European & African.

2. Platform popularity

- PS2 has the largest aggregate reported `total_sales` among platforms

3. Top-selling game records

- The highest sold individual game per platform is *Grand Theft Auto V* on PS3, at **20.32 million**.

Evaluation, Limitations and Conclusion

What worked well

- Pandas provides a consistent process for cleaning and transforming the data.
- Interactive Plotly charts allow users to explore several business questions.
- The CRISP-DM process provides a structured approach from business understanding to evaluation.

Data limitations

- The original dataset contains 64,016 rows.
- Approximately **70.4% of `total_sales` values are missing**.
- Approximately **80.3% of `na_sales` and 89.5% of `jp_sales` values are missing**.
- Missing data can change rankings, regional comparisons and apparent trends.

Possible improvements

- Investigate the possibility to recover some missing sales values.
- Improve the dashboard with better data representation.
- Reassess conclusions after improving data quality.

END